

MAJOR ASSIGNMENT

Q.01) The weight of students of different classes are given below:

Weights	Classes		
(Kilograms)	A	B	C
$20 \leq x < 30$	7	5	6
$30 \leq x < 40$	10	9	25
$40 \leq x < 50$	20	21	24
$50 \leq x < 60$	18	15	4
$60 \leq x < 70$	7	6	3

Compute that which group is more consistent.

Q.02) Given the following data of income per week of employees and number of Earners as:

Income per week (Rs)	41 - 50	51 - 60	61 - 70	71 - 80	81 - 90	91 - 100
No. of Earners	30	36	43	104	73	14

Compute the following:

- Inter and semi-inter Quartile Range. Also compute its coefficient.
- Mean Deviation about Mode and its coefficient

Q.03) The following data shows the traffic density and accident rate of Peshawar city.

Traffic Density	30	35	40	45	50	60	70	80	90
Accident Rate	2	4	5	5	8	15	24	30	32

- Compute the correlation coefficient between Traffic Density and Accident Rate. Also interpret the result.
- Fit the simple linear regression model Traffic Density on Accident Rate. Also interpret the Model.
- Estimate Traffic Density, when Accident rate is 40.